

Building a CPU from Scratch

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From bits to brains(CPU).

Objective

The goal of this project is to design and implement a processor in Verilog through three stages:

1. Single-Cycle CPU(week-5)
2. Multi-Cycle CPU
3. Pipelined CPU

The final processor should be capable of executing a Bubble Sort program on an array stored in memory and demonstrate the performance differences between the three architectures.

My Progress:

W1 – Verilog Basics

Studied the fundamentals of Verilog language using HDL bits, like creating gates, wires including Modules and ports, Continuous assignments, Procedural blocks, Testbench creation, Simulation workflow using Icarus Verilog and GTKWave and also installing quartus..

W2 – Digital Components

Prepared the following cpu's building blocks:

Arithmetic Logic Unit (ALU)

Implemented arithmetic and logical operations such as:

- Addition
- Subtraction
- AND
- OR
- Shift operations

Multiplexers(mux 2:1, mux 4:1)

Implemented and tested the above mux codes with the given test benches...

Register File

Implemented a register file supporting read and write operations.

Program Counter (PC)

Built and tested a basic Program Counter module.

Week 3 – Finite State Machines(studied from Brown & Vranesic, 3rd Ed)

Studied how finite state machines are built...

Topics covered:

- Moore machines.
- Mealy machines.
- State transitions when there are finite number of states.
- And also the types of resets: synchronous and asynchronous.

And for assignment I did lemmings problems where our code is used for execution of lemmings game... It is inspired from the idea of how a processor can switch states when the clk or reset are there. The same analogy is used for the lemmings..

Week 4 – CPU Architecture and Assembly

Studied processor organization and instruction execution.

And for assignment I designed opcode decoder and its test bench and implemented it.

Topics covered:

- Datapath design
- Control unit
- Opcode decoding
- Instruction execution cycle
- Assembly language fundamentals
- i281 ISA overview

Gained understanding of how instructions flow through the processor and how control signals leads to execution, which was helpful for next week's execution of bubblesort algorithm.....

Current Status

The fundamental processor components have been studied and implemented individually.

Modules completed:

- ALU
- Program Counter
- Register File
- Multiplexers
- Opcode Decoder

Need to design data memory, code memory which are required for executing bubble sort algo of single cycle cpu.

Plan for Single-Cycle CPU Implementation} bcz I need to complete it..

- Integrate datapath components.
- Implement code memory.
- Implement data memory.
- Execute assembly programs on the processor. And also use i281 simulator.
- Implement Bubble Sort on an array stored in memory.

What I had learnt so far:

Through this project I have gained practical exposure to:

- ✓ Hardware Description Languages (Verilog)
- ✓ Digital logic design
- ✓ Processor architecture
- ✓ Datapath and control design
- ✓ Simulation and debugging workflows

Seasons of Code 2026 – Midterm Report

These topics have helped build a stronger understanding of how modern processors are designed and operate internally.

